

Elgamal Cryptosystem

Elgamal Facts:

- Public Key system based on the discrete logarithm problem, $y = g^x \bmod p$.
- Developed in 1984/1985 by T.Elgamal
- Used in the digital signature standard (DSS).
- Uses a random integer k in the creation of the ciphertext so that a constant plaintext will result in a different ciphertext every time it's encrypted.
- Has a ciphertext that is double the size of the plaintext.

"A" has public key (p, g, y) , where p is a large prime, g is a primitive root, and $y = g^x \bmod p$.

The secret key, x , is difficult to compute for large values of p .

To send a message m to "A", where $0 \leq m \leq p-1$:

- Get a random integer k , where $1 < k < p-1$
- Compute: $c_1 = g^k \bmod p$
- Compute: $c_2 = m(y^k) \bmod p$
- Send ciphertext $c = (c_1, c_2)$

"A" decrypts $c = (c_1, c_2)$ by computing $c_2 [c_1^{-x}] \bmod p$.

$$\begin{aligned} c_2 [c_1^{-x}] \bmod p &= m(y^k) [(g^k)^{-x}] \\ &= m(y^k) [g^{-xk}] \\ &= m((g^x)^k) [g^{-xk}] \\ &= mg^{xk} [g^{-xk}] \\ &= m \end{aligned}$$

Example:

"A" has public key $(p = 19, g = 10, y = 3)$, and secret key $x = 5$.

To send a message $m = 17$ to "A":

- Get a random integer, let's say we get $k = 6$
- Compute: $c_1 = 10^6 \bmod 19 = 11$
- Compute: $c_2 = 17(3^6) \bmod 19 = 5$
- Send ciphertext $c = (11, 5)$

"A" decrypts $c = (11, 5)$ by computing $5[11^{-5}] \bmod 19$:

First, get the inverse of $11 \bmod 19$, which is 7.

$$\begin{aligned} 5[11^{-5}] \bmod 19 &= 5[(11^{-1})^5] \bmod 19 \\ &= 5[(7)^5] \bmod 19 \\ &= 17 \end{aligned}$$

Note: if we resend this message with a different random k we'll get a different ciphertext:

To re-send the message $m = 17$:

- Get another random integer, let's say we get $k = 4$
- Compute: $c_1 = 10^4 \bmod 19 = 6$
- Compute: $c_2 = 17(3^4) \bmod 19 = 9$
- Send ciphertext $c = (6, 9) \neq (11, 5)$

"A" decrypts $c = (6, 9)$ by computing $9[6^{-5}] \bmod 19$:

First, get the inverse of $6 \bmod 19$, which is 16.

$$\begin{aligned} 9[6^{-5}] \bmod 19 &= 9[(6^{-1})^5] \bmod 19 \\ &= 9[(16)^5] \bmod 19 \\ &= 17 \end{aligned}$$

Exercise:

Akuma has Elgamal public key $(p = 13, g = 6, y = 3)$.

1. Send the message $m = 2$ to *Akuma* using the random integer $k = 10$.
2. Determine *Akuma*'s secret key x .
3. Use the secret key x to decrypt the ciphertext from question 1.

Answers:

1. $c = (4, 6)$

2. $x = 8$

3. $6(4^{-8}) \bmod 13 = 6(10)^8 \bmod 13 = 2$